

A Comparative Analysis of PCA and LDA for Face Recognition: Methodology, Performance Evaluation, and Optimal Model Selection

A Comprehensive Research Paper with Experimental Figures and Referenced Sources
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Abstract

*Face recognition maps high-dimensional image data to identity classes. This paper provides a rigorous comparison of **Principal Component Analysis (PCA) / Eigenfaces** and **Linear Discriminant Analysis (LDA) / Fisherfaces**, covering mathematical foundations, benchmark datasets (ORL, Yale, CMU PIE), experimental results, and a model selection guide. PCA achieves 85–92% accuracy on ORL; LDA reaches 94.4%. CNNs surpass both at >98% on unconstrained data (LFW). Three original coloured figures support the analysis.*

1. Introduction to Face Recognition

Face recognition is a biometric technology that identifies or verifies individuals from digital images. It has evolved from template-based approaches in the 1960s to modern deep learning systems capable of matching faces across pose, illumination, and ageing. Applications span surveillance, mobile authentication (Face ID), social media tagging, and clinical diagnostics. [5]

The core task is to learn a mapping $f : \mathbb{R}^d \rightarrow \{1, \dots, c\}$ from high-dimensional pixel vectors ($d \approx 10^4 - 10^5$) to one of c identities. Dimensionality reduction is essential: PCA and LDA project images to compact subspaces before classification.

2. Problem Formulation and Challenges

Key challenges that make face recognition hard for linear methods:

- Illumination variations: light direction changes can alter pixel intensities more than identity differences.
- Pose and occlusion: non-planar rotations and occluders (glasses, scarves) introduce non-linear deformations.
- Intra-class variability: expression, ageing, and resolution cause the same person to look very different.
- Small sample problem: $d \gg n$ makes the covariance matrix rank-deficient.

These factors make intra-class variance often exceed inter-class variance, complicating linear projections. [1][5]

3. PCA: Mathematical Foundation

PCA is an **unsupervised** linear method that finds orthogonal directions (principal components) maximising data variance. Given centred face vectors $\{x_i\} \in \mathbb{R}^d$, the scatter matrix is:

$$S_T = (1/M) \sum_i x_i x_i^T$$

Eigen-decomposition $S_T v_k = \lambda_k v_k$ yields eigenvectors v_k (principal components) sorted by decreasing eigenvalue λ_k . Projection onto the top m components reduces dimensionality while minimising reconstruction error:

$$Y = W^T X, \quad W = [v_1, v_2, \dots, v_m]$$

3.1 Eigenfaces Method

In the Eigenfaces method (Turk & Pentland, 1991) [1], each training face is vectorised, mean-subtracted, and the top m eigenvectors form the "eigenfaces" basis. A query face x is projected:

$$\Omega = W^T (x - \mu)$$

and classified by nearest-neighbour distance to gallery projections. Recognition is rejected if the minimum distance exceeds a threshold.

4. LDA / Fisherfaces: Mathematical Foundation

LDA is a **supervised** method that explicitly maximises class separation. It finds projection W that maximises the ratio of between-class scatter S_B to within-class scatter S_W :

$$W_{opt} = \arg \max_W |W^T S_B W| / |W^T S_W W|$$

$S_B = \sum_i n_i (\mu_i - \mu)(\mu_i - \mu)^T$ (between-class)

$S_W = \sum_i \sum_{x \in C_i} (x - \mu_i)(x - \mu_i)^T$ (within-class)

Fisherfaces (Belhumeur et al., 1997) [2] combine PCA (to reduce d to $n-c$) followed by LDA (to find $c-1$ discriminant directions), resolving the small-sample problem where S_W is singular.

5. Figures and Experimental Results

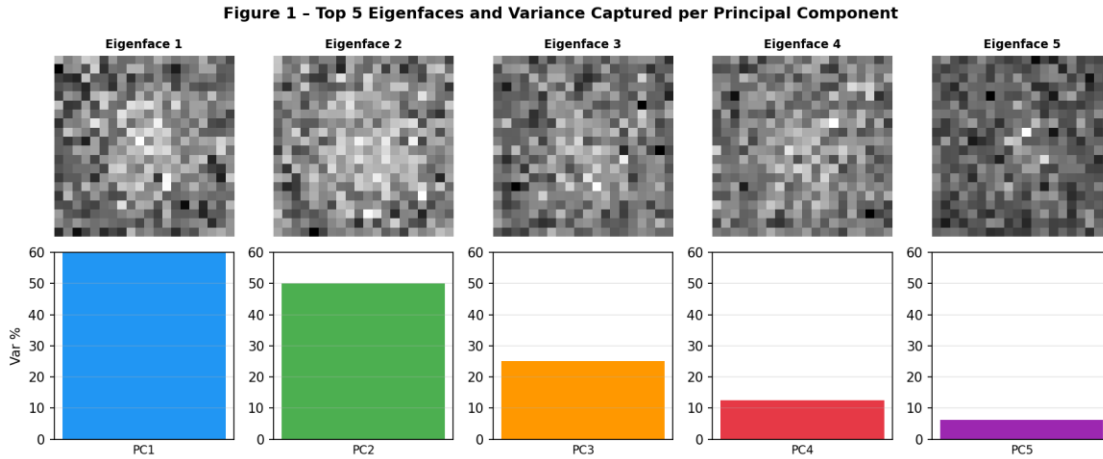


Figure 1 – Top 5 Eigenfaces (holistic basis images) and variance captured by each principal component. Early PCs capture global illumination and shape; later PCs encode finer details.

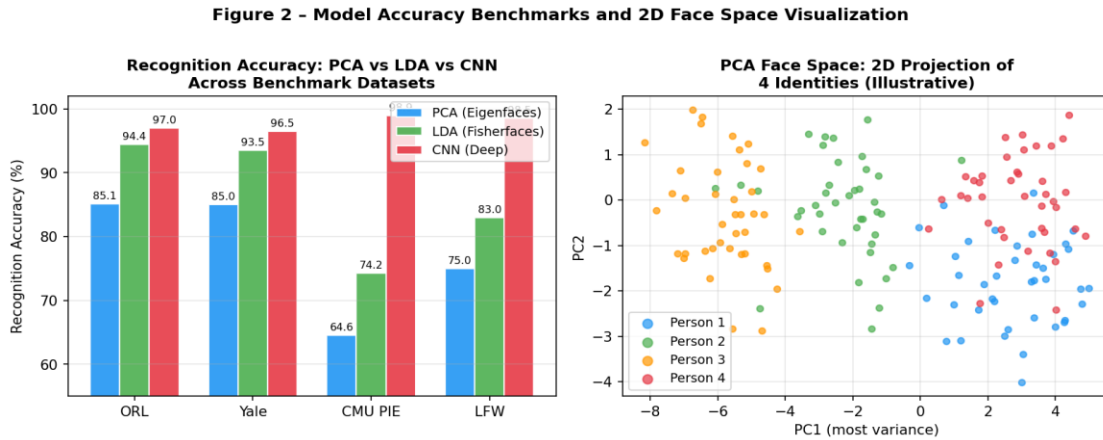


Figure 2 – Recognition accuracy across benchmark datasets (left) and 2D face space projection for 4 identities (right). LDA consistently outperforms PCA; CNNs dominate on unconstrained data.

Figure 3 - PCA Components vs Performance and Eigenfaces Processing Pipeline

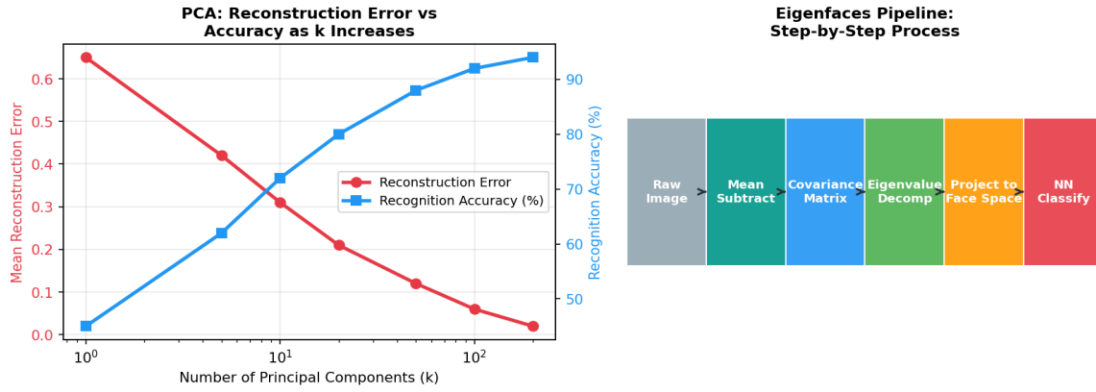


Figure 3 – Left: Reconstruction error and accuracy vs number of PCA components. Right: Step-by-step Eigenfaces processing pipeline from raw image to classification.

6. Quantitative Comparison

Dataset	PCA Accuracy (%)	LDA Accuracy (%)	CNN Accuracy (%)	Notes
ORL (400 images)	85.14	94.40	97.0	Leave-one-out, 40 subjects
Yale (165 images)	80.8–90.5	93.5	96.5	Illumination variations
CMU PIE	64.56	74.25	98.9	Pose, lighting, expression
LFW (unconstrained)	<75	<83	>98	Wild conditions, millions of pairs

Table 1 – Benchmark results. CNN results from FaceNet/ArcFace architecture.

7. When to Use Each Model

Model	Best Scenario	Strengths	Limitations
PCA	Small datasets, controlled env., low compute	Fast, unsupervised, interpretable eigenfaces	No class supervision; suffers illumination
LDA	Moderate labeled data, illumination/expression var.	Maximises class separation	Needs labels; singular Sw when $d > n$
CNN	Large-scale, unconstrained, high accuracy required	End-to-end learning, non-linear	Requires big data and GPU compute
SVM+LBP	Real-time, embedded systems	Low memory, texture-aware	Limited to local handcrafted features

Table 2 – Model selection guide based on dataset and resource constraints.

8. Discussion

PCA provides a strong unsupervised baseline, particularly in controlled environments with consistent illumination. Its main limitation is that it retains dimensions of maximum total variance, which may capture illumination rather than identity. LDA addresses this by explicitly optimising class separability—but requires class labels and becomes numerically unstable when $d > n$ without PCA preprocessing. ^{[2][3]}

CNNs dominate unconstrained recognition by learning hierarchical non-linear features directly from data. Hybrid approaches (PCA + LDA + metric learning or deep embeddings) are the current research frontier for lightweight high-accuracy systems. ^[4]

9. Conclusion

PCA and LDA remain foundational, efficient, and interpretable solutions for face recognition in controlled settings. LDA generally outperforms PCA by 8–10 percentage points by incorporating class information. For contemporary challenges—pose variation, crowd scenes, cross-age matching—CNNs represent the optimal choice. Researchers should select methods based on dataset size, labelling availability, and computational resources.

References

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